

Deepesh Rathod

Founding Engineer · Lead Architect · Crisis EM

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Founding Engineer with six years from first line of code to founding-team scale: ~2 years self-directed personal projects (2020 → mid-2021), then ~4.5 years professional — **3 internships** across early-stage startups, then **Founding Engineer at Zoca** (Sep 2022 → May 2026).

Defined the foundational AWS ECS/Fargate stack that scaled Zoca from 90 to **1,500+ customers (\$3.5M ARR, 140+ headcount)**. Architected the **\$6M Lambda Swarm** Step Functions hierarchy that closed the Series A. Shipped the Master Stroke — **Scheduling & Payments v2** (Stripe Connect + tenant isolation across 1,500+ tenants + booking snapshots) — in **30 days** on \$200/mo Claude Max with a small high-agency team. Published **17 ADRs · 13 case studies · 7 long-form doctrines**.

Looking for the next big build — **Founding Engineer / Staff Software Engineer / Tech Lead** roles where the work crosses system seams.

DEFINING WORK

- **Stack-defining decisions:** chose AWS ECS + Fargate over EC2; consolidated 3 servers + 50–60 Lambdas into **6 autoscaling services + 10 workers + 10 Lambdas**; orchestrated zero-failure migration of all legacy users — the calls that supported scale from 90 → 1,500+ customers (\$1M+ → \$3.5M ARR).
- **\$6M Lambda Swarm:** hierarchical Step Functions + S3 JSON state — the Streamlit Revenue Opportunity engine that closed the Series A. **2-hour batch → 5-minute pipeline**.
- **AI-native engineering platform:** Claude Code + custom Metabase MCP + AWS CLI toolchain that compresses outage and customer-issue RCAs from **2–3 hours → ~5 minutes** (25–35× speedup); auto-updating doc pipeline and Mac-based local infra clone enabling AI agents to write tests and validate end-to-end flows without staging dependencies.
- **Master Stroke (Feb 2026):** shipped **Scheduling & Payments v2** — Stripe Connect + tenant isolation across 1,500+ tenants + booking-item snapshots — in **30 days** on \$200/mo Claude Max with a 3-engineer team.
- **Force-multiplier leadership:** helped scale engineering from 2–3 → 15–18; ran **20+ technical interviews**; **Senior Buddy** for 4+ engineers (mentees shipping bug fixes in week 1 and end-to-end features within their first month); owned audit infrastructure and AWS ownership after senior departures.

EXPERIENCE

Zoca — Founding Member · Lead Architect · Crisis EM

Jan 2025 — May 2026

AWS ECS/Fargate · Terraform · Drizzle ORM · Redis · NestJS · Stripe Connect · LLM infra · SOC2/HIPAA

- Architected the foundational stack for a multi-tenant SaaS that scaled from **90 → 1,500+ customers** (1,000+ paying, **\$1M+ → \$3.5M ARR**, 140+ headcount); chose AWS ECS+Fargate over EC2 and consolidated 3 servers + 50–60 Lambdas into 6 autoscaling services + 10 workers + 10 Lambdas, drastically reducing operational surface while preserving pod-driven delivery.
- Designed **tenant isolation + database-layer audit logging** — every internal access gated through a unified RBAC system and recorded as immutable change history; foundation for **SOC2/HIPAA readiness**.
- Led **zero-failure migration** of all legacy users to the new architecture — no data loss, no production incidents, no rollbacks.
- Built an **AI-native engineering toolchain** (Claude Code + custom Metabase MCP + AWS CLI) compressing RCAs from 2–3 hours of dedicated engineering bandwidth to ~5 minutes — complemented by an auto-updating doc pipeline and a Mac-based local infra clone for AI-driven E2E testing.
- Authored the **Master-Worker pattern** (Redis + NestJS workers + Lambda callbacks), the **LLM Guardrails** safety layer, the **Drizzle ORM + Terraform IaC** rollout for SOC2/HIPAA-readable change management, and the **Win Agent's deterministic LLM testing** suite (invariants over examples).
- Led the **5-Minute RCA** on the Infra-Flip CI/CD seam; authored the *System Wisdom vs Framework Knowledge* doctrine.
- Integrated **Retell / Voice AI** for an early experimental cohort and built the observability layer — surfacing latency, error, and behavioral signals the moment each call ends.
- Led the Scheduling & Payments Pod — RFC processes, code-review standards, architectural guidelines; ran 20+ technical interviews; Senior Buddy for 4+ engineers.
- **Master Stroke (Feb 2026)**: shipped **Scheduling & Payments v2** — Stripe Connect + tenant isolation + booking-item snapshots — in **30 days** with a 3-engineer team on \$200/mo Claude Max.
- Built a local **Slack PR Reviewer** and an **Agentic Docs Pipeline** keeping ~80% of owned codebase docs in sync with commits — Engineering-Manager-as-Builder responses to rising AI velocity.

Zoca — Founding Engineer

Jun 2023 — Dec 2024

Node.js · NestJS · AWS (Lambda, Step Functions, S3, ECS/Fargate) · PostgreSQL · 150 GB media migration

- Spearheaded migration of the mobile-app backend from a legacy Python monolith to **Node.js / NestJS services**; established team-wide DevEx standards and shared utility tooling that scaled cleanly with the team.
- Engineered a **distributed media pipeline** handling 1,000+ user uploads/day across **900+ GB of S3** with content-aware duplicate detection (0.1–5% reclaimed); executed the 150 GB migration with 30-hour scripts, two crashes, pause/resume + caffeinate — **0% data loss**.
- Owned the **Flask → Node fallback migration** with a client-side safety net (zero-failure cutover for live users).
- Drove SQL performance work: **15-minute query → 75 seconds** via CTEs, composite indexes, deadlock resolution.
- Owned the entire **AWS Cloud Infrastructure & DevOps** after a senior-leadership transition — absorbed full infra ownership while keeping product delivery uninterrupted.
- Architected a **hyper-local keyword demand system** with 5-mile-radius geo-accuracy by integrating US Census data.
- Designed and shipped **Scheduling v1** end-to-end — slot algorithm, transaction blocks, entity-snapshot pattern (the "Integrity Engine").
- Architected the **\$6M Lambda Swarm** — hierarchical Step Functions + S3 JSON state; the Streamlit Revenue Opportunity engine that closed the \$6M round.
- Authored the *Cynical Architect / Recovery-First* doctrine and the **PSQL Audit-by-Snapshot** pattern.

Zoca — Software Engineer Intern

Sep 2022 — May 2023

Python · Jinja · lemmatization · image classification · PostgreSQL · SQL triggers

- Built **citation and NAP (Name/Address/Phone) consistency automation** — delivered a 1-month scoped initiative in **1 week**, freeing the founding team for higher-leverage work.
- Implemented **lemmatization-based keyword matching** to improve search relevance and recall.
- Built a **dynamic website generation pipeline** combining Jinja templates with image classification.
- Led a **1.5-hour Christmas-morning P0** (Dec 25, 2022) — production rescue of a deleted-row incident.
- Authored the **SQL-Defined Business Triggers** pattern — AFTER-DELETE audit triggers on every core table; zero silently-lost rows since.

Arbunize Digital (VideoWiki) — Backend Developer Intern

Jan 2022 — Apr 2022

Python · Django · FFmpeg · real-time progress tracking

- Built backend APIs in **Python/Django** using FFmpeg for the VideoWiki video editor.
- Engineered a **real-time progress tracking system** for long-running FFmpeg tasks, improving editor UX.

Yosecal Tech LLP (Finmo) — SDE Intern

Oct 2021 — Dec 2021

NestJS · AWS SES · CloudTrail · SQS · Slack API · transactional email

- Built NestJS servers integrating **AWS SES, CloudTrail, SQS, and Slack API** — powering management-event logging and transactional email rendering.

PERSONAL PROJECTS · 2020 — MID 2021 (SELF-DIRECTED FULL-STACK FOUNDATION)

Quiz App

React · Node.js · MongoDB Atlas · Passport · JWT · Redux · MaterialUI · Axios

Full-stack web app with user login/registration, Passport authentication, JWT authorization, Redux global state, Mongoose data layer. github.com/deepesh-01/quiz-app

News-App

React · Node.js · Axios · News API · Heroku-deployed

Production-style React frontend with a Node middleware connecting to a third-party news API; deployed on Heroku.

Instagram Automation

Python · Selenium

Selenium-driven Chrome automation — early exposure to browser-driven automation patterns that scaled into the Master Stroke's \$200/mo Claude Max workflow five years later.

The IPO framework (*Input → Process → Data Context → Output*) was first drafted during this period as personal engineering doctrine; later became the documented backbone for every blueprint and ADR at Zoca.

SELECTED PUBLISHED DOCTRINES · 17 ADRS AT V1.DEEPESH-ENGG.IN

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|-----------------|--|
| ADR-0009 | The Lambda Swarm — Step Functions hierarchy + S3 JSON state · \$6M Series A closed |
| ADR-0010 | Recovery-First Architecture — the Cynical Architect doctrine |
| ADR-0011 | Audit-by-Snapshot — generic PSQL before/after logging |
| ADR-0012 | Master-Worker Pattern — Redis + NestJS workers + Lambda callbacks |
| ADR-0014 | Drizzle ORM + Terraform IaC for SOC2/HIPAA-readable infrastructure |
| ADR-0015 | Deterministic LLM Testing — invariants over examples |
| ADR-0016 | System Wisdom vs Framework Knowledge — the Founding Logic doctrine |

WEEKEND / OFF-HOURS ARCHITECTURAL EXPLORATION

Slack PR Reviewer (Claude + Slack + Git): LLM-powered sentiment + technical review on PRs via emoji reaction — reviewed 30 PRs in the first 3 days vs ~15 baseline manually (**2× review throughput** with improved consistency).

Agentic Docs Pipeline: AI agents that keep ~80% of an owned codebase's docs in sync with commits; the end-state of the AI-Native Handover doctrine.

WeLog: collaborative real-time journaling for partners; end-to-end full-stack (auth, signup, encrypted private data, real-time shared entries with per-user isolation).

MarketSignal AI: real-time trading-automation engine ingesting Angel One market data on a 15-min cron, evaluating conditions, delivering Telegram alerts — full pipeline on local infrastructure, no third-party orchestration.

Private Cloud on Raspberry Pi: self-hosted server stack via Cloudflare Tunnels (no public IPs, no port-forwarding) — **2+ months continuous uptime, 3–5s cold boot**; engineered a passwordless camera-to-laptop streaming flow over unified LAN.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL (PostgreSQL), Bash, HTML, CSS, C++, Java, C
Backend / Runtime	Node.js, NestJS, Express, Flask, FastAPI, Django, Django REST Framework, Streamlit
Frontend	React, React Native, Redux, MaterialUI
Cloud / Infra	AWS (ECS, Fargate, EC2, Lambda, Step Functions, S3, IAM, RDS, SES, SQS, CloudTrail, CloudFront), Terraform, Docker, CI/CD, GitHub Actions, Cloudflare Pages, Cloudflare Tunnels
Data / Storage	PostgreSQL (CTEs, triggers, RLS, transactions, indexes, optimisation), Redis, MongoDB, Drizzle ORM, audit-by-snapshot logging
AI / LLM	Claude Code, Claude Max, OpenAI Function Calling, Anthropic Claude API, custom MCP servers (Metabase MCP), LLM guardrails, deterministic LLM testing, semantic serialization, AI agent observability, Retell / Voice AI
Architecture	Multi-tenant SaaS, microservices, monorepo, tenant isolation + RBAC, zero-downtime migrations, master-worker pattern, Stripe Connect, scheduling / slot algorithms, booking-item snapshots
Compliance / Method	SOC2 readiness, HIPAA readiness, RBAC, audit logging, change management, Architecture Decision Records (ADR), RFC processes, IPO framework
Leadership	Crisis Engineering Manager, Tech Lead, founding-engineer cohort, hiring & technical interviewing, Senior Buddy mentorship, code review & doctrine authorship

EDUCATION

B.Tech — Information Technology · Shri Shankaracharya Technical Campus, SSGI (CSVTU affiliated) · 2019 — 2023 · GPA 7+/10

Full portfolio + 17 ADRs + 13 case studies: v1.deepesh-engg.in